Name: ______________________ Partner: ____________________ Period: ________

The Tangent Function — Tiered Practice

The tangent function connects the unit circle to slope and to periodic asymptotic behavior. Choose **one tier** below based on your readiness level. Everyone should be prepared to explain their reasoning to the class.

Tier R — Remediate

Context: The unit circle values below are given. Use them to compute $\tan\theta = \sin\theta/\cos\theta$ at each angle.

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
0	0	1	0
$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3} = \sqrt{3}/3 \approx 0.577$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3} \approx 1.732$
$\pi/2$	1	0	**undefined**
$3\pi/4$	$\sqrt{2}/2$	$-\sqrt{2}/2$	-1
π	0	-1	0

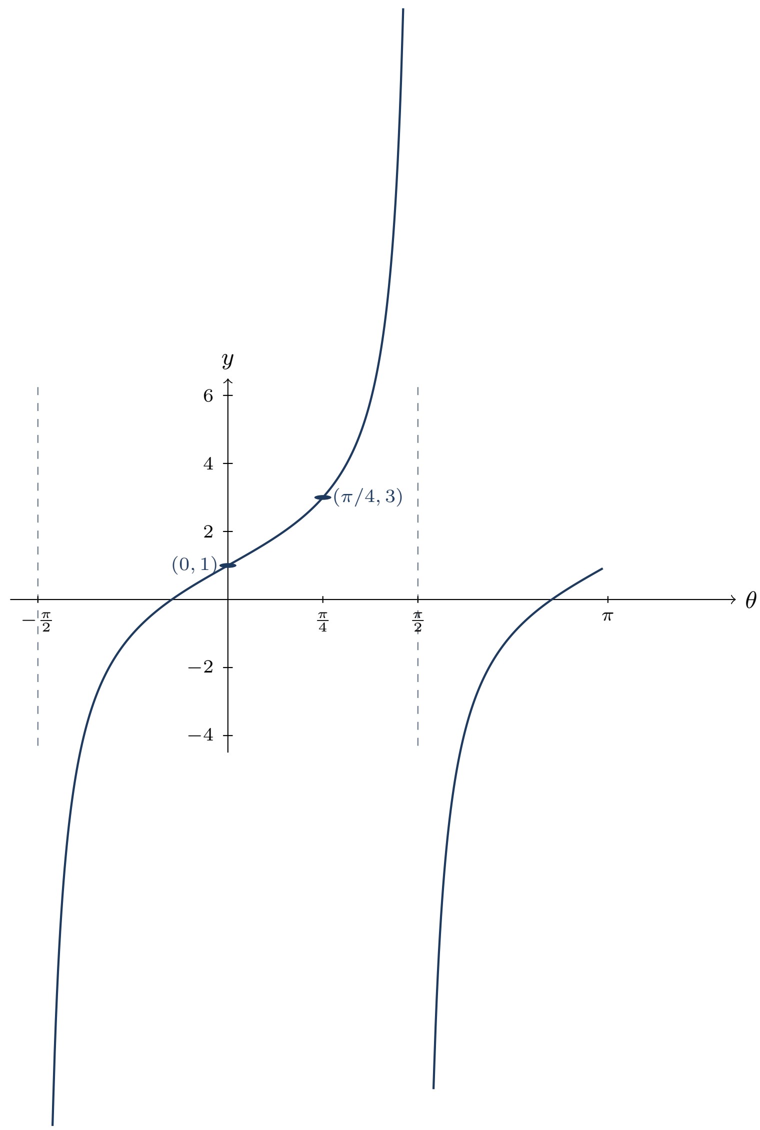
1. At which angles above is $\tan\theta$ *undefined*? Why?
$\tan\theta$ **is undefined at** $\theta = \pi/2$ **because** $\cos(\pi/2) = 0$ **and division by zero is undefined.**

2. The tangent function has vertical asymptotes at $\theta = \pi/2 + k\pi$ for integer k. List the three asymptotes closest to $\theta = 0$.
$\theta = -\pi/2$**,** $\theta = \pi/2$**,** $\theta = 3\pi/2$

3. Between $\theta = 0$ and $\theta = \pi/2$, is $\tan\theta$ increasing or decreasing? How do you know from the table?
Increasing: the values go $0, \sqrt{3}/3, 1, \sqrt{3}, \ldots$ **— each output is larger than the previous one as** θ **increases.**

Tier A — Approaching Proficiency

Context: The graph below shows one and a half periods of a tangent function $f(\theta) = a\tan(\theta) + d$.



1. What is the period of f?

π (the distance between consecutive asymptotes is π)

2. List all vertical asymptotes visible in the graph.

$\theta = -\pi/2$, $\theta = \pi/2$, $\theta = 3\pi/2$

3. On the interval $(-\pi/2, \pi/2)$, is f concave up or concave down for $\theta < 0$? For $\theta > 0$?

Concave down for $\theta < 0$; concave up for $\theta > 0$. The curve bends downward (like a hill) to the left of the origin and upward (like a bowl) to the right.

4. Find a and d.

At $\theta = 0$: $f(0) = a\tan(0) + d = d = 1$, so $d = 1$.

At $\theta = \pi/4$: $f(\pi/4) = a \cdot 1 + 1 = 3$, so $a = 2$.

Therefore $f(\theta) = 2\tan(\theta) + 1$.

Tier E — Extension

Context: Analyze $g(\theta) = 2\tan\left(\dfrac{\theta}{2} - \dfrac{\pi}{4}\right) + 1$.

1. Rewrite in the form $a\tan(b(\theta + c)) + d$.

Factor: $\dfrac{\theta}{2} - \dfrac{\pi}{4} = \dfrac{1}{2}\left(\theta - \dfrac{\pi}{2}\right)$**.**

So $g(\theta) = 2\tan\left(\frac{1}{2}(\theta - \frac{\pi}{2})\right) + 1$**.**

$a = 2$, $b = \frac{1}{2}$, $c = -\frac{\pi}{2}$, $d = 1$.

2. Period of g:

Period $= \pi/|b| = \pi \div (1/2) = 2\pi$**.**

3. Vertical asymptotes:

Set $\frac{1}{2}(\theta - \frac{\pi}{2}) = \frac{\pi}{2} + k\pi$**:**

$\theta - \frac{\pi}{2} = \pi + 2k\pi$

$\theta = \frac{3\pi}{2} + 2k\pi$ **for integer** k**.**

So asymptotes at $\dots, -\frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{2}, \dots$

4. Compute $g(0)$ and $g(\pi/2)$:

$g(0) = 2\tan\left(-\frac{\pi}{4}\right) + 1 = 2(-1) + 1 = -1$.

$g(\pi/2) = 2\tan\left(\frac{\pi}{4} - \frac{\pi}{4}\right) + 1 = 2\tan(0) + 1 = 2(0) + 1 = 1$.

5. Description of transformations:

Compared to $y = \tan\theta$**: the graph is vertically dilated by a factor of 2 (outputs doubled), has period** 2π **instead of** π **(horizontal stretch by factor 2), shifted right by** $\pi/2$ **(phase shift), and shifted up by 1 unit (vertical translation).**